

psql Cheat Sheet

PP2 — Programming Principles 2 | PostgreSQL command-line reference

How to enter psql: Windows: `psql -U postgres` | macOS: `psql postgres` | Linux: `sudo -i -u postgres && psql` — prompt becomes `postgres=#`

Connecting

CONNEC
T

psql -U postgres

Connect as postgres user (from terminal)

CONNEC
T

psql -U user -d dbname

Connect to a specific database (terminal)

CONNEC
T

psql -U user -d db -h localhost -p 5432

Connect with explicit host and port

CONNEC
T

\c dbname

Switch to a different database (inside psql)

CONNEC
T

\c dbname username

Switch database as a different user

CONNEC
T

\conninfo

Show current connection info

CONNEC
T

\q

Quit / exit psql

Navigating databases & tables

NAVIGAT
E

\l

List all databases

NAVIGAT
E

\l+

List databases with size and owner info

NAVIGAT
E

\dt

List all tables in current database

NAVIGAT
E

\d tablename

Show columns, types, constraints of a table

NAVIGAT
E

\d+ tablename

Detailed table info (storage, triggers)

NAVIGAT
E

\dn

List all schemas

NAVIGAT
E

\di

List all indexes

NAVIGAT
E

\dv

List all views

NAVIGAT
E

\ds

List all sequences

NAVIGAT
E

\df

List all functions

NAVIGAT
E

\du

List all users and roles

Creating & modifying tables (DDL)

DDL

CREATE DATABASE mydb;

Create a new database

DDL

DROP DATABASE mydb;

Delete a database permanently

DDL

**CREATE TABLE students (
id SERIAL PRIMARY KEY,
name VARCHAR(100) NOT NULL,
grade INT
);**

Create a table with auto-increment id

DDL

DROP TABLE students;

Delete a table

DDL

DROP TABLE IF EXISTS students;

Delete a table only if it exists (safe)

DDL

**ALTER TABLE students ADD COLUMN email
TEXT;**

Add a new column

DDL

ALTER TABLE students DROP COLUMN email;

Remove a column

DDL

**ALTER TABLE students RENAME COLUMN
grade TO score;**

Rename a column

DDL

**ALTER TABLE students ALTER COLUMN name
TYPE VARCHAR(200);**

Change a column's data type

Inserting, updating & deleting (DML)

DML

**INSERT INTO students (name, grade)
VALUES ('Ali', 90);**

Insert a single row

DML

**INSERT INTO students (name, grade)
VALUES
('Ase1', 85), ('Dau1et', 92);**

Insert multiple rows at once

DML

**UPDATE students SET grade = 95
WHERE name = 'Ali';**

Update rows matching a condition

DML

UPDATE students SET grade = grade + 5;

Update ALL rows — no WHERE = all affected!

DML

DELETE FROM students WHERE grade < 50;

Delete rows matching a condition

DML

DELETE FROM students;

Delete ALL rows (table structure remains)

DML

TRUNCATE TABLE students;

Faster delete, resets sequences

Querying data

QUERY `SELECT * FROM students;`

Select all rows and columns

QUERY `SELECT name, grade FROM students;`

Select specific columns

QUERY `SELECT * FROM students WHERE grade > 80;`

Filter with a condition

QUERY `SELECT * FROM students ORDER BY grade DESC;`

Sort results descending

QUERY `SELECT * FROM students LIMIT 10;`

Get first 10 rows only

QUERY `SELECT * FROM students LIMIT 10 OFFSET 20;`

Pagination: skip 20, take next 10

QUERY `SELECT COUNT(*) FROM students;`

Count total rows

QUERY `SELECT AVG(grade), MAX(grade), MIN(grade) FROM students;`

Aggregate functions

QUERY `SELECT grade, COUNT(*) FROM students GROUP BY grade;`

Group rows by a column value

QUERY `SELECT * FROM students WHERE name LIKE 'A%';`

Pattern match — names starting with A

QUERY `SELECT s.name, c.title FROM students s JOIN courses c ON s.id = c.student_id;`

Inner join between two tables

QUERY `SELECT DISTINCT grade FROM students;`

Get unique values only

Users & permissions (Admin)

ADMIN `CREATE USER alice WITH PASSWORD 'secret';`

Create a new user

ADMIN `DROP USER alice;`

Delete a user

ADMIN `ALTER USER alice WITH PASSWORD 'newpass';`

Change a user's password

ADMIN `ALTER USER alice CREATEDB;`

Allow user to create databases

ADMIN `GRANT ALL PRIVILEGES ON DATABASE mydb TO alice;`

Full access to a database

ADMIN `GRANT SELECT, INSERT ON students TO alice;`

Grant specific table permissions

ADMIN `REVOKE INSERT ON students FROM alice;`

Remove a specific permission

ADMIN `\timing`

Toggle query execution time display

ADMIN `\i /path/to/file.sql`

Run SQL commands from an external file

ADMIN `\o /path/to/output.txt`

Redirect query output to a file

ADMIN `\e`

Open last query in system text editor